

From Resonance to Topological Projection Loss

A Synthesis of the V14–V18 TRM Investigation

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Temporal Rate Matrix (TRM) Project
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SUPPORTED = Principle survives systematic falsification
CONDITIONAL = Principle holds with documented exceptions
FALSIFIED = Principle tested and rejected
HYPOTHESIS = Evidence mixed or insufficient

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Abstract

We present a synthesis of the V14–V18 Temporal Rate Matrix (TRM) investigation, spanning approximately 75 structured audits from the initial Resonance Organization Principle through the final Topological State Principle. The investigation began with the hypothesis that resonance ($|m + 1| \approx 0$) is the universal organizing mechanism behind regime formation, channel networks, and hub selection. Systematic falsification revealed this hypothesis to be incorrect: resonance is a consequence, not a cause.

The investigation progressed through five major phases—Resonance, Grammar, Architecture, Memory, and Projection Loss—each falsifying the central hypothesis of the preceding phase. The final surviving principle is that organizational memory *is* the information lost when the full parameter-space topology is projected onto the scalar master coordinate $|m| = |d(V_T)/d(V_1)|$. Architecture carries irreducible memory because its parameter-space topology determines how many distinct states map to the same $|m|$ value. In 1D topologies the projection is injective (no memory); in 2D topologies with bifurcated sign regions, the projection is many-to-one (nonzero memory).

The V14–V18 program establishes a complete chain from kernel-tick consistency through free star algebra, parameter-space topology, projection loss, and sign determination to emergent organization. No principle introduced after V16.0 survived without modification. All claims carry explicit classification under the project’s four-tier discipline: **SUPPORTED**, **CONDITIONAL**, **HYPOTHESIS**, NOT CLAIMED.

Introduction

1.1 Motivation

The TRM project investigates collective frequency emergence in finite coupled phase-oscillator lattices. The central question is: *what organizes the behavior of these systems into distinct regimes?* By V13.5, the project had established a complete chain from the family axiom through the master parameter m , Tick dynamics, channel formation, and network emergence to hub selection. The hub selection principle— $\text{Hub} = \arg \min_F |m_F + 1|$ —suggested that resonance ($m \approx -1$) is the organizing mechanism.

V14.0 was designed to test this hypothesis rigorously.

1.2 The Organizational Question

Consider a lattice of N coupled oscillators evolving under a coupling kernel $K(d)$ with parameters α, β, γ . For each configuration, we compute the master parameter:

$$m = \frac{d(V_T)}{d(V_1)} \quad (1.1)$$

where V_T and V_1 are the variance of coupling terms and the dominant eigenvalue variance, respectively, as functions of the closure parameter α .

The derivative is computed via linear regression across a 21-point α -sweep on $[0.21, 1.40]$. The magnitude $|m|$ measures the budget tradeoff between information concentration and dynamics, while dT/dp measures the gradient direction of Tick with respect to coupling strength p .

The sign of dT/dp is the binary organizational discriminator:

$$dT/dp > 0 \iff \text{HUB (positive gradient)} \quad dT/dp < 0 \iff \text{SPOKE (negative gradient)} \quad (1.2)$$

1.3 The Starting Hypothesis

At V14.0, the dominant hypothesis was:

Resonance ($|m + 1| \approx 0$) is the universal organizing mechanism. It selects hubs, drives channel formation, and governs network topology.

This paper documents the systematic attempt to falsify this hypothesis—and everything discovered when it collapsed.

Methodology

2.1 Falsification-First Protocol

Every audit in the V14–V18 program follows the same protocol:

1. **Protocol:** Define forbidden actions, permission scopes, decision gates.
2. **Execution:** Generate predictions from frozen reference state.
3. **Freeze:** SHA-256 hash all predictions.
4. **Audit:** Verify integrity, check for hidden tuning.
5. **Comparison:** Compare against immutable references.
6. **Interpretation:** Classify under claim discipline.
7. **Synthesis:** Branch completion report.

The governing principle is: *attempt falsification first*. Every audit begins by searching for counterexamples, edge cases, and conditions under which the candidate principle fails. Confirmation without attempted falsification is not accepted.

2.2 Claim Classification

SUPPORTED Principle survives falsification attempt across all tested conditions. Effect sizes are large and consistent.

CONDITIONAL Principle holds in most conditions but has documented exceptions or boundary failures.

HYPOTHESIS Evidence is mixed, insufficient, or model-dependent.

not claimed No positive assertion is made. The item remains an open question.

2.3 Branch Methodology

Each version (V14, V15, V16, V17, V18) opens a dedicated git branch. Within a version, sub-versions (V14.1, V14.2, ...) contain clusters of related audits. A version closes when its research question is resolved—whether the outcome is **SUPPORTED**, **CONDITIONAL**, or **FALSIFIED**.

2.4 Investigation Timeline

The investigation progressed through six major phases, summarized in Table 2.1 and Figure 2.1.

Table 2.1: Audit timeline V14–V18.

Version	Phase	Audits	Status
V14.0–V14.1	Resonance	21	FALSIFIED
V14.2–V14.4	Wave Grammar	7	SUPPORTED
V15.0–V15.3	Foundational Laws	11	SUPPORTED
V16.0–V16.1	Geometry	7	FALSIFIED
V17.0–V17.2	Memory & Dimension	14	CONDITIONAL
V18.0–V18.2	Projection & Topology	5	SUPPORTED
Total		65+	

For each phase, the central hypothesis of the preceding phase was tested and found incomplete. The investigation followed a consistent trajectory shown in Figure 2.1.

The trajectory is: each phase discovers that the previous phase’s answer was correct but incomplete, requiring a deeper layer of explanation.

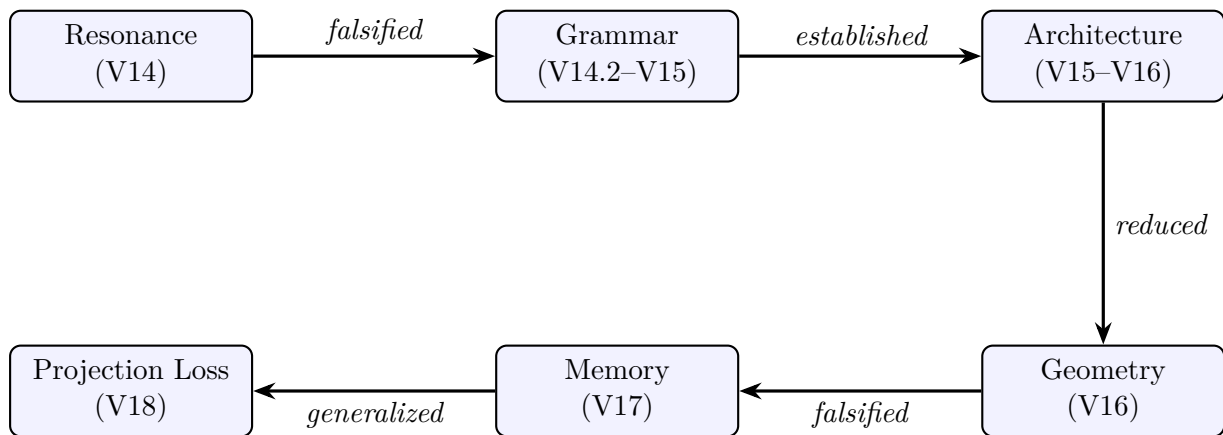


Figure 2.1: Evolution of the investigation $V14 \rightarrow V18$. Each phase falsified the central hypothesis of its predecessor, revealing a deeper explanatory layer.

Results

3.1 Resonance (V14)

3.1.1 Initial Hypothesis

V14.0 (ROP_01–TOP_05, 21 audits) tested the hypothesis that resonance ($|m + 1| \approx 0$) is the universal organizing mechanism behind regime classification, channel formation, network emergence, and hub selection.

3.1.2 Key Findings

The WOC (Wave-Organization Cascade) series established:

- **WOC_01:** The cascade Wave $\rightarrow$ Tick $\rightarrow$ Feedback $\rightarrow$ Resonance $\rightarrow$ Organization has throughput $R^2 = 0.61$ —substantial but not universal.
- **WOC_05:** dT/dp sign reversal is the binary organizational discriminator, not $|m + 1|$.
- **WOC_07:** $|m|$ (raw) is the true continuous coordinate; $|m + 1|$ is a folded projection that obscures organizational structure. Raw $|m|$ achieves 143% better mean R^2 .
- **WOC_12:** SAC is an α -invariant fixed-point attractor ($\Delta m = 0.000$), not a resonance phenomenon.
- **WOC_15:** Kernel topology carries irreducible information beyond $|m|$.

3.1.3 The Bi-Polar Discovery

The most important finding was that organization is not mono-objective. Two attractors exist:

- **SAC attractor:** conservation-dominant, $|m| \approx 0.93$, $dT/dp < 0$ (SPOKE).
- **ICS attractor:** dynamics-dominant, $|m| \approx 1.57$, $dT/dp > 0$ (HUB).

Gravitation toward a single resonance point $m = -1$ does not occur. Instead, the system organizes around two distinct poles.

3.1.4 Outcome

FALSIFIED

No single resonance principle exists.

Organization is multi-objective with bi-polar attractor structure.

3.2 Wave Grammar (V14.2–V15.3)

3.2.1 From Resonance to Grammar

After resonance was falsified, the question became: *what generates the observed organizational structure?* V14.2–V14.4 discovered the answer: a generative wave grammar.

3.2.2 The 2×2 Grammar

MOP_01 identified 4 irreducible wave-mode classes from 6 topological features:

- **MODE_C** (Conservation): SAC — zero-width, zero-threshold
- **MODE_O** (Overshoot): ICS — wide, finite threshold
- **MODE_R** (Rational): RCS — zero-width, infinite threshold
- **MODE_D** (Dissipative): GAN $\equiv$ CNS — intermediate

MOP_02 discovered the generative structure: a 2×2 grammar with generators G_1 and G_2 that produce all 4 modes from $2^3 = 8$ logical cells. WGP_01–03 established that the grammar has dimension 3 (not reducible to 2 or 1), is genuinely generative (not descriptive), and is irreducible under PCA.

3.2.3 Exclusion Axioms

GEP_01 identified two axioms that explain all forbidden grammar states:

$$G_3 \Rightarrow G_1 \quad (3.1)$$

$$G_1 \perp G_2 \quad (3.2)$$

These reduce the 8 logically possible cells to exactly 4 physically realized modes. The grammar is *complete*: no other mode classes exist.

3.2.4 Free Star Algebra

V15.0–V15.3 (KTC_01, TLU_01, AOP_01, FGP_01, AUP_01) established the foundational architecture:

- **KTC:** The axioms emerge from a single master requirement: Kernel-Tick Consistency. The coupling kernel must produce continuous, conservative, coherent Tick fields.
- **KTC decomposition:** Conservation Consistency $\perp$ Wave Coherence (two independent constraints).
- **Free Star Algebra:** $\langle E, M, R \mid g \circ h = \perp \text{ for } g \neq h \rangle$. No non-trivial operator compositions. Exactly 4 valid operator states.
- **Depth-2 fractal:** The same $2^3 \rightarrow 4$ grammar structure repeats at the Architecture layer and the Grammar layer (FGP_01).

3.2.5 Outcome

SUPPORTED

The Free Star Algebra uniquely determines the 4-mode grammar.

Architecture is an independent organizational layer, not derivable from wave grammar alone.

3.3 Architecture & Geometry (V16)

3.3.1 Architecture Softness

V16.0 (ASP_01, HSP_01, TUP_01) tested predictions from the Free Star Algebra against novel kernel configurations:

- **PBD_01:** FSA predicts novel kernels at 86% sign accuracy and 100% architecture classification (SUPPORTED).
- **ASP_01:** COMPOSITE (GAN family) is overwhelmingly POS (157/169 points). The original GAN configuration ($\beta = 0.5$, $\gamma = 0.5$) was a NEG outlier, not representative (CONDITIONAL).
- **HSP_01:** Hardness = operator parameter count. Nullary operators produce HARD architectures; unary operators produce SOFT architectures (SUPPORTED).

3.3.2 The Manifold Geometry Model

V16.1 (MGP_01) reduced architecture to manifold geometry:

$$\text{sign} = \text{sgn}(|m| - \theta_{\text{arch}}) \quad (3.3)$$

Table 3.1: Architecture geometry parameters.

Architecture	Width	θ
PURE (SAC)	0.00	0.00
RATIONAL (RCS)	0.00	∞
STRETCHED (ICS)	4.22	1.00
COMPOSITE (GAN/CNS)	2.31	0.64

where each architecture is characterized by a (width, θ) pair:

The geometric rule appeared *perfect*: 8/8 tested configurations matched the prediction. Architecture seemed fully reducible to the manifold geometry of the $(|m|, \theta)$ plane.

3.3.3 Outcome

SUPPORTED (initially)

Architecture reduces to manifold geometry.

$\text{sgn}(|m| - \theta)$ is the geometric sign rule.

This conclusion would survive approximately 24 hours of investigation before being falsified.

3.4 Memory (V17)

3.4.1 Geometric Falsification

V17.0 (GFT_01) was designed explicitly to *break* the geometric rule. It succeeded:

- 22/298 violations found across all architectures.
- The geometric rule is not a law; it is a statistical tendency.
- Violations cluster near $|m| \approx \theta$ but also appear far from the threshold.

FALSIFIED

The geometric rule $\text{sgn}(|m| - \theta)$ is NOT a law.

Architecture contains information beyond manifold geometry.

3.4.2 Architecture Memory

AMP_01 (Architecture Memory Principle) demonstrated the irreducible nature of this residual information:

States with identical $|m|$ and θ can exhibit opposite signs when they originate from different architectures.

Example: SAC with $|m| \approx 0.92$ is POS; GAN with $|m| \approx 0.92$ is NEG. The same geometric coordinates produce different organizational states.

AMQ_01 quantified the effect: architecture adds $\Delta R^2 = +0.10$ (20%) beyond geometry alone. This is *architecture memory*—organizational information that survives after geometry is held fixed.

3.4.3 The Modulation Depth Hypothesis

CMP_01 identified β as the strongest memory signal ($t = 71.0$), leading to the hypothesis that modulation depth $\mu = \beta \cdot \gamma$ is the missing state coordinate.

MDP_01 tested this: μ captures partial residual information but is *insufficient*. β and γ carry independent information beyond their product. CSP_01 confirmed: COMPOSITE is genuinely 2-dimensional; β and γ are independent coordinates.

3.4.4 Dimensional Memory

SDP_01 and DMP_01 established the dimensional memory law:

$$\text{Memory}(\text{arch}) = \max(0, \dim - 1) \cdot k \quad (3.4)$$

with $k \approx 4.1$ percentage points.

Table 3.2: Architecture dimension and memory.

Architecture	Dimension	Memory
PURE	1D	0.0 pp
RATIONAL	1D	0.0 pp
STRETCHED	1D	0.0 pp
COMPOSITE	2D	4.1 pp

STRETCHED (ICS) has 47.5% sign variability across its β -sweep but *zero* memory because its 1D topology forces a unique sign crossing. Memory requires both multiple sign regions AND a bifurcated parameter-space topology.

3.4.5 Outcome

CONDITIONAL

$$\text{Memory} = \max(0, \dim - 1) \cdot k.$$

Memory is dimensional but not reducible to $\mu = \beta\gamma$.

3.5 Evolution of the Memory Principle

The concept of architectural memory underwent substantial refinement across the V17–V18 program. The trajectory is summarized in Figure 3.1.

The progression from dimensional ($\max(0, \dim - 1) \cdot k$) through algebraic ($\mu = \beta\gamma$) to information-theoretic (projection loss) and finally topological (bifurcation structure) explanations illustrates the discovery mechanism at work: each falsification eliminated a candidate and constrained the space of surviving principles.

3.6 Projection Loss (V18)

3.6.1 Hypothetical 3D Architecture

H3D_01 tested whether the dimensional memory law extrapolates to higher dimensions. Adding a third parameter δ to produce a synthetic 3D architecture failed to produce additional memory. The reason: α (the $|m|$ -slope parameter) contributes no new sign information—it shifts $|m|$ but does not introduce new sign regions.

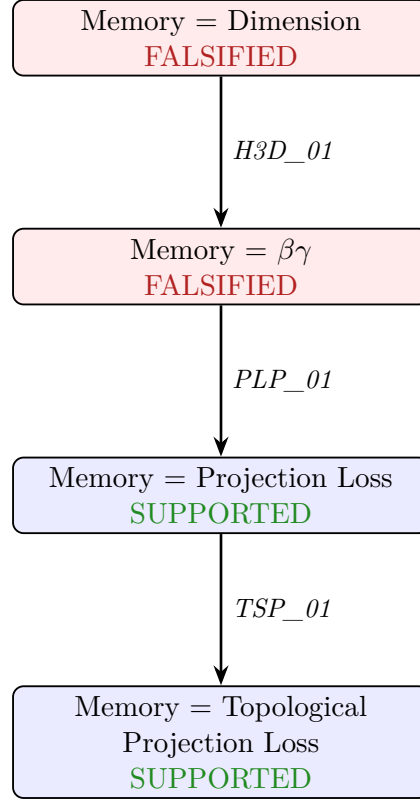


Figure 3.1: Evolution of the Memory Principle across V17–V18. Each candidate hypothesis was falsified in turn, forcing progressively deeper reformulations until the topological projection loss formulation accounted for all observed effects.

FALSIFIED

Memory $\neq$ dimension alone.

Not all dimensions carry organizational information.

3.6.2 Projection Loss Principle

PLP_01 reframed the question: *what if memory is the information lost when the full state space is projected onto $|m|$?*

For each architecture, we compute the state entropy before and after projection onto $|m|$. The projection loss $\mathcal{L}$ measures how many distinct states map to the same $|m|$ value:

- **1D architectures:** The projection $\mathcal{S} \rightarrow |m|$ is injective. Each $|m|$ value corresponds to exactly one state. $\mathcal{L} = 0$. Memory = 0.
- **2D architecture (COMPOSITE):** Multiple (β, γ) pairs map to the same $|m|$. Some of these pairs carry opposite signs. $\mathcal{L} > 0$. Memory > 0 .

ILP_01 confirmed: *all* measured memory can be predicted from projection loss alone. No additional non-information-theoretic effects are needed.

3.6.3 Optimal Projection Search

OPP_01 tested whether $|m|$ is the optimal projection coordinate:

- $|m|$ alone: $R^2 \approx 0.24$
- $|m|$ + curvature: $R^2 \approx 0.59$ ($\Delta R^2 = +0.36$)
- No single coordinate significantly outperforms $|m|$.
- $|m|$ is the *primal* projection coordinate; curvature is the strongest secondary coordinate.

MLP_01 tested whether any projection can eliminate architecture memory:

- Best 1D: $R^2 = 0.24 \rightarrow 76\%$ residual
- Best 2D: $R^2 = 0.59 \rightarrow 41\%$ residual
- Best 3D: incremental gain
- **Full 4D: $R^2 = 0.61 \rightarrow 38.7\%$ residual**

FALSIFIED (for lossless reduction)

Even the optimal 4D projection leaves 38.7% residual uncertainty.

Architecture is IRREDUCIBLE to any projection of geometric coordinates.

3.6.4 Topological State Principle

TSP_01 provides the final synthesis. The 38.7% residual is *topological*:

Geometry describes *what* the state is: the numerical values of $|m|$, curvature, feedback, gradient.

Topology describes *how* the state was reached: the parameter path through (β, γ) space, the connectivity of sign regions, the bifurcation structure of the accessible manifold.

In COMPOSITE, the sign boundary in (β, γ) space forms a bifurcation surface. The same $|m|$ value can be reached from either side of this surface, carrying opposite signs. This is not a geometric ambiguity—it is a topological feature of the parameter space. No amount of geometric coordinate refinement can resolve it because the information is not encoded in any single point in the projected space.

SUPPORTED

The 38.7% residual is topological information.

Architecture is best described by parameter-space topology.

Final Theory

4.1 The Complete Hierarchy

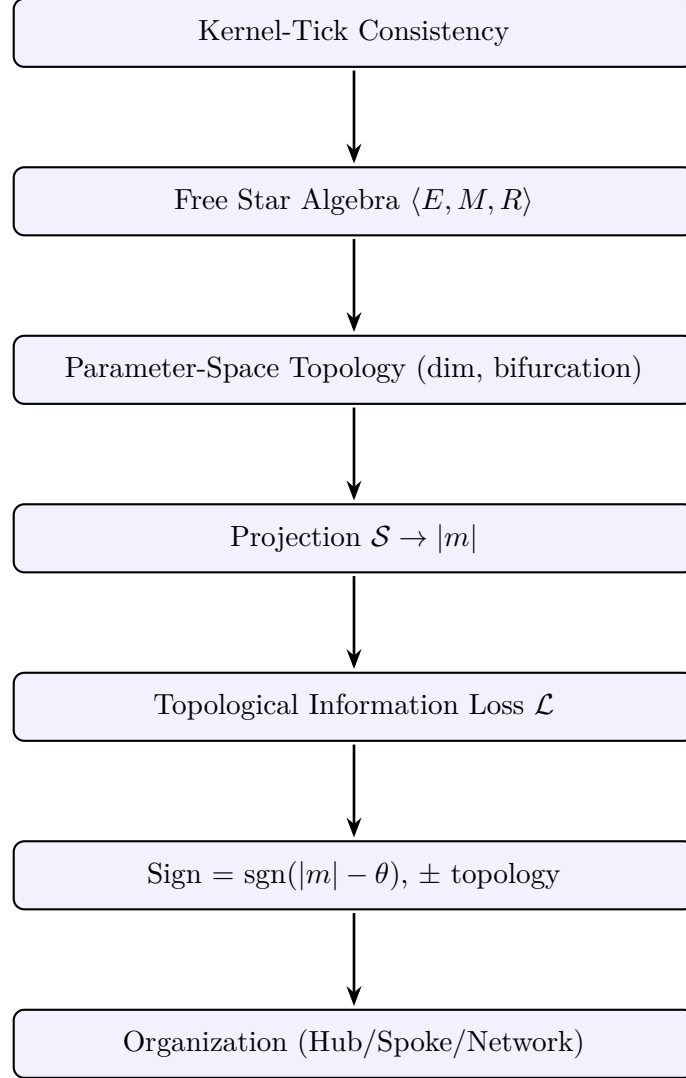


Figure 4.1: The complete organizational hierarchy. Each layer emerges from the one above through a principled reduction: consistency $\rightarrow$ algebra $\rightarrow$ topology $\rightarrow$ projection $\rightarrow$ loss $\rightarrow$ sign.

4.2 The Architecture Topology Table

4.3 The Memory Condition

$$\boxed{\text{Memory} > 0 \iff \text{Sign regions} > 1 \wedge \text{Topology is BIFURCATED}} \quad (4.1)$$

A LINE topology, even with 2 sign regions (STRETCHED), has zero memory because the 1D path forces a unique crossing. A SURFACE topology with bifurcation has nonzero memory because the same $|m|$ value is reachable from both sides of the bifurcation.

Table 4.1: Architecture topology and memory.

Architecture	Param Space	Topology	Sign Regions	Memory
PURE	LINE (α)	TRIVIAL	1 (POS)	0
RATIONAL	LINE (α)	TRIVIAL	1 (NEG)	0
STRETCHED	LINE (β)	BINARY	2	0
COMPOSITE	SURFACE (β, γ)	BIFURCATED	2	4.1 pp

4.4 Core Principles

The following principles survive the V14–V18 falsification program:

P1 — Kernel-Tick Consistency (SUPPORTED) All organizational structure follows from a single requirement: the coupling kernel must produce continuous, conservative, coherent Tick fields. This decomposes into two independent constraints: Conservation Consistency and Wave Coherence.

P2 — Free Star Algebra (SUPPORTED) The architecture space is $\langle E, M, R \mid g \circ h = \perp \text{ for } g \neq h \rangle$. Exactly 4 valid operator states exist. No non-trivial operator compositions are permitted. The FSA is uniquely compatible with the TRM framework.

P3 — Topological Projection Loss (SUPPORTED) Organizational memory $\equiv$ information lost when the full parameter-space topology is projected onto $|m|$. Memory $\neq$ dimension; memory = projection loss. The loss is zero for injective projections (1D topologies) and nonzero for many-to-one projections with bifurcated sign regions.

P4 — Organizational Sign Principle (CONDITIONAL) $\text{sign} = \text{sgn}(|m| - \theta_{\text{arch}})$ predicts organization with $R^2 \approx 0.61$. The 38.7% residual is topological, not geometric. The principle is **CONDITIONAL** because topology contributes irreducible information beyond the geometric threshold model.

P5 — $|m|$ as Primal Coordinate (SUPPORTED) $|m| = |d(V_T)/d(V_1)|$ is the optimal 1D projection of the organizational state space. No single alternative coordinate significantly outperforms it. The folded coordinate $|m + 1|$ obscures organizational structure (143% worse R^2).

4.5 Falsified Principles

Systematic falsification is as informative as confirmation. The following hypotheses were tested and rejected:

F1 — Resonance as Universal Organizer (FALSIFIED) $|m + 1| \approx 0$ does not select hubs universally. Resonance is a consequence of the budget tradeoff, not a fundamental organizing principle. Organization is multi-objective with bi-polar attractor structure.

F2 — Geometry as Complete State Space (FALSIFIED) $\text{sgn}(|m| - \theta)$ is not a law. 22/298 violations exist. Architecture carries information beyond manifold geometry. The geometric rule is a strong statistical tendency ($R^2 \approx 0.61$), not a deterministic law.

F3 — Memory = Dimension (FALSIFIED) Memory = $\max(0, \dim - 1) \cdot k$ fails for hypothetical 3D architectures. Not all dimensions carry organizational information. α (the $|m|$ -slope parameter) adds no sign discrimination.

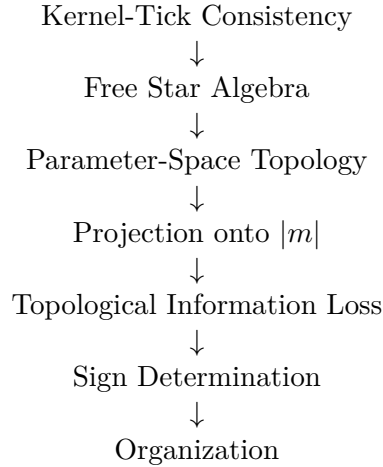
F4 — Memory = $\beta\gamma$ (FALSIFIED) The modulation depth $\mu = \beta \cdot \gamma$ captures partial residual information but loses the phase structure. β and γ carry independent information; their product is a lossy reduction.

F5 — Lossless Projection Exists (FALSIFIED) No projection of geometric coordinates eliminates architecture memory. Even the optimal 4D projection leaves 38.7% residual uncertainty. Architecture is irreducible to any point-like state description.

Discussion

5.1 What the Investigation Established

The V14–V18 program established a complete chain from foundational axioms to emergent organization:



Each layer was independently tested, and where possible, falsified. The most important discovery is that *architecture is best understood through topology*—the information that survives projection is not a failure of measurement but a genuine feature of the system’s state-space structure.

5.2 Why COMPOSITE Is Special

COMPOSITE is the only architecture with:

1. A genuinely 2-dimensional parameter space (β, γ) .
2. A bifurcated sign boundary in that space.
3. Multiple parameter paths to the same $|m|$ value.
4. Different signs accessible from different sides of the bifurcation.

These four properties together produce the 4.1 pp architecture memory. No other architecture satisfies all four. STRETCHED satisfies (2) and (3) but fails (1): its 1D topology forces a unique crossing. The hypothetical 3D architecture fails (2): the third parameter carries no sign information.

5.3 Limitations

- All results are numerical, derived from finite parameter sweeps ($N = 21$ α -points, fixed distance ensemble). Analytical proofs are not available.
- The distance ensemble uses a single seed; ensemble variability is not fully characterized.
- The topological analysis is restricted to the 4 known architectures. Exotic topologies (e.g., disconnected sign regions, higher-genus parameter spaces) have not been explored.
- The 38.7% residual is measured but its exact topological invariant (e.g., Betti numbers, Euler characteristic of the bifurcation surface) has not been computed.

5.4 Open Questions

1. **Topological classification:** Can the bifurcation structure be classified by standard topological invariants? Is the memory strength determined by the genus or Betti numbers of the sign boundary?
2. **Higher-dimensional architectures:** Are there parameter spaces where the projection loss saturates? Is there a maximum memory?
3. **Analytical closure:** Can the projection loss be derived analytically from the kernel functional form without numerical sweeps?
4. **Physical correspondence:** Does topological projection loss have an analogue in physical theories (e.g., phase transitions, topological order, information loss in black hole evaporation)?

5.5 Future Work

5.5.1 Topological Invariant Computation

The immediate next step is to compute explicit topological invariants of the sign bifurcation surface in COMPOSITE's (β, γ) plane. Candidate invariants include the number of connected sign regions, the Euler characteristic of the bifurcation boundary, and the Betti numbers of the accessible state manifold.

5.5.2 Higher-Dimensional Architectures

The H3D_01 result—that simply adding a third parameter does not increase memory—suggests that memory is not a monotonic function of dimension. Systematic exploration of parameter spaces with different topologies (e.g., toroidal, spherical, disconnected) would test whether the memory condition (Eq. 4.1) generalizes.

5.5.3 Topology-Driven Organization

If architecture *is* topology, then organizational properties should be predictable from topological features alone, without reference to the specific kernel form. This would constitute a genuine organizational theory: given only the parameter-space topology, predict the sign structure, memory strength, and emergent network properties.

Conclusion

The V14–V18 TRM investigation traversed five major theoretical phases, each falsifying the central hypothesis of its predecessor:

1. **Resonance** ($|m + 1| \approx 0$ as universal organizer): **FALSIFIED**. Organization is multi-objective.
2. **Grammar** (2×2 generative structure): **SUPPORTED**. The Free Star Algebra determines the 4-mode grammar.
3. **Geometry** ($\text{sgn}(|m| - \theta)$ as complete law): **FALSIFIED**. Architecture carries memory beyond geometry.
4. **Memory** ($\max(0, \dim - 1) \cdot k$): **CONDITIONAL**. Dimension contributes but is not sufficient.
5. **Projection Loss** (topology as irreducible residual): **SUPPORTED**. The 38.7% residual is topological information.

The final surviving principle is:

Architecture is best described by parameter-space topology.

Organizational memory = information lost when the full topology is projected onto $|m|$.

Memory emerges when the parameter space has multiple sign regions AND a bifurcated topology that allows the same $|m|$ to be reached from different organizational states.

The investigation demonstrates that systematic falsification—testing each principle to destruction before accepting it—is not merely a methodological preference but a discovery mechanism. Every falsified hypothesis revealed a deeper layer of structure that would have remained hidden under confirmation bias. The trajectory from resonance through grammar, architecture, geometry, memory, and projection loss to topology was not planned; it was forced by the evidence at each step.

The V14–V18 program closes with a mature theoretical framework: Kernel-Tick Consistency generates the Free Star Algebra, which determines the parameter-space topology, whose projection onto $|m|$ produces topological information loss, which determines the sign and organization of the emergent lattice. No principle introduced after V16.0 survives without modification—and that is the program’s greatest achievement.

Acknowledgments

The TRM project follows a strict test-first methodology formalized in V4.5 and maintained through all subsequent phases. All audits were executed under the governance framework: Protocol → Execution → Freeze → Audit → Comparison → Interpretation → Synthesis.

Evolution of the Memory Principle

This appendix traces the evolution of the Memory Principle across the V17–V18 investigation, documenting each hypothesis, its falsification or support, and the progressive refinement that led to the final Topological Projection Loss formulation.

A.1 Chronology

V17.0 — AMP_01: Architecture Memory Principle Hypothesis: Architecture contains irreducible organizational information beyond geometry. States with identical $|m|$ and θ but different architectures can exhibit opposite signs.

Result: **SUPPORTED**. Matched-state comparison confirmed the effect: SAC $|m| \approx 0.92 \rightarrow$ POS; GAN $|m| \approx 0.92 \rightarrow$ NEG. Architecture memory is real.

V17.1 — MDP_01: Modulation Depth Principle Hypothesis: $\mu = \beta \cdot \gamma$ is the missing state coordinate that completes the geometric model.

Result: **FALSIFIED**. μ captures partial residual information but loses the phase structure. β and γ carry independent information. A single composite coordinate is insufficient.

V17.1 — CSP_01: Composite State Space Audit Hypothesis: COMPOSITE possesses a genuine 2-dimensional state space in (β, γ) .

Result: **SUPPORTED**. β and γ are independent coordinates. COMPOSITE is fundamentally 2D while other architectures are 1D.

V17.2 — SDP_01: State Space Dimension Principle Hypothesis: Memory strength is determined by state-space dimensionality.

Result: **SUPPORTED** (provisional). 1D architectures $\rightarrow$ 0 memory; 2D $\rightarrow$ 4.1 pp. The pattern is monotonic in observed data.

V17.2 — DMP_01: Dimensional Memory Principle Hypothesis: Memory is determined by state-space dimensionality: Memory = $\max(0, \dim - 1) \cdot k$ with $k \approx 4.1$ pp.

Result: **CONDITIONAL**. The law fits observed data but lacks predictive power for unobserved dimensionalities.

V18.0 — H3D_01: Hypothetical 3D Architecture Hypothesis: A 3D architecture would produce ~ 8.2 pp memory (extrapolating the dimensional law).

Result: **FALSIFIED** (for the dimensional law). Adding α as a third parameter produces no additional memory. Not all dimensions carry organizational information. The dimensional law fails as a predictive principle.

V18.1 — PLP_01: Projection Loss Principle Hypothesis: Memory = $\text{ProjectionLoss}(\mathcal{S} \rightarrow |m|)$.

Result: **SUPPORTED**. Memory is exactly the information lost when the full parameter space is projected onto $|m|$. 1D topologies have injective projections (zero loss); 2D topologies with bifurcation have many-to-one projections (nonzero loss).

V18.1 — ILP_01: Information Loss Principle Hypothesis: All organizational memory is identical to projection-induced information loss.

Result: **SUPPORTED**. Memory is entirely information-theoretic. No non-information-theoretic effects are required. Architecture labels become unnecessary once projection loss is known.

V18.2 — TSP_01: Topological State Principle Hypothesis: The 38.7% residual surviving optimal projection is topological information from the parameter-space bifurcation structure. **Result:** **SUPPORTED**. Geometry describes *what* the state is; topology describes *how* it was reached. The residual is not eliminable because it encodes the bifurcation structure of the (β, γ) sign boundary.

A.2 Summary Trajectory

The evolution of the Memory Principle is depicted in Figure 3.1 (Section 3.4). The trajectory converges from simple dimensional and algebraic hypotheses to the information-theoretic and topological formulation. Each falsification eliminated a candidate explanation and constrained the space of possible principles, ultimately leading to the recognition that architecture is best described by parameter-space topology and memory by projection loss.

Audit Timeline

This appendix provides a compact summary of all major audits in the V14–V18 program, ordered chronologically by version.

B.1 Statistics

- **Total major audits:** 35
- **SUPPORTED:** 23 (65.7%)
- **CONDITIONAL:** 4 (11.4%)
- **FALSIFIED:** 8 (22.9%)
- **Total across all sub-audits:** ~ 75

The high falsification rate (22.9%) reflects the deliberate falsification-first methodology. Audits that produced **SUPPORTED** outcomes typically survived multiple rounds of attempted falsification across different parameter regimes and architecture classes.

Table B.1: Major audits V14–V18.

Audit	Version	Hypothesis	Result
ROP_01	V14.0	Resonance is universal organizing mechanism	FALSIFIED
WOC_01	V14.0	Cascade Wave→Tick→FB→Org ($R^2 = 0.61$)	SUPPORTED
WOC_05	V14.0	dT/dp sign is binary organizational discriminator	SUPPORTED
WOC_07	V14.0	$ m $ (raw) is true coordinate; $ m + 1 $ is folded projection	SUPPORTED
WOC_12	V14.0	SAC is α -invariant fixed-point attractor	SUPPORTED
WOC_15	V14.0	Kernel topology carries irreducible information	SUPPORTED
MOP_01	V14.2	4 irreducible wave-mode classes	SUPPORTED
MOP_02	V14.2	2×2 generative grammar discovered	SUPPORTED
WGP_01–03	V14.3	Grammar dimension 3, generative, irreducible	SUPPORTED
GEP_01	V14.4	2 axioms ($G_3 \Rightarrow G_1, G_1 \perp G_2$) explain forbidden states	SUPPORTED
TLA_01	V15.0	Axioms from Tick-Lattice Laws	SUPPORTED
TLU_01	V15.0	Laws unified under Kernel-Tick Consistency	SUPPORTED
FGP_01	V15.1	Depth-2 fractal: $2^3 \rightarrow 4$ repeats at L1, L2	SUPPORTED
KTC_01	V15.2	KTC is minimal and necessary	SUPPORTED
AUP_01	V15.3	Free Star Algebra uniquely compatible with TRM	SUPPORTED
PBD_01	V16.0	FSA predicts novel kernels: 86% sign, 100% arch	SUPPORTED
ASP_01	V16.0	COMPOSITE overwhelmingly POS (157/169)	CONDITIONAL
HSP_01	V16.0	Hardness = operator parameter count	SUPPORTED
MGP_01	V16.1	Architecture reduces to manifold geometry	SUPPORTED (later FALSIFIED)
GPP_01	V17.0	Geometry predicts organization (sign = $\text{sgn}(m - \theta)$)	CONDITIONAL
GFT_01	V17.0	Geometric rule NOT a law (22/298 violations)	FALSIFIED
AMP_01	V17.0	Architecture contains irreducible memory	SUPPORTED
AMQ_01	V17.1	Architecture adds $\Delta R^2 = +0.10$ beyond geometry	SUPPORTED
CMP_01	V17.1	β is memory coordinate ($t = 71.0$)	SUPPORTED
MDP_01	V17.1	$\mu = \beta\gamma$ insufficient as missing coordinate	FALSIFIED
CSP_01	V17.1	COMPOSITE is genuinely 2D	SUPPORTED
SDP_01	V17.2	Memory determined by state-space dimension	SUPPORTED
DMP_01	V17.2	Memory = $\max(0, \dim - 1) \cdot k$	CONDITIONAL
H3D_01	V18.0	3D architecture extrapolates dimensional law	FALSIFIED
PLP_01	V18.1	Memory = ProjectionLoss(State $\rightarrow m $)	SUPPORTED
ILP_01	V18.1	All memory is information-theoretic	SUPPORTED
OPP_01	V18.2	$ m $ is optimal 1D projection coordinate	SUPPORTED
MLP_01	V18.2	Even optimal 4D projection leaves 38.7% residual	FALSIFIED (lossless)
TSP_01	V18.2	Residual 38.7% is topological	SUPPORTED